

Laboratory Work 1: ERD Diagram

Topic: International Airport Management System

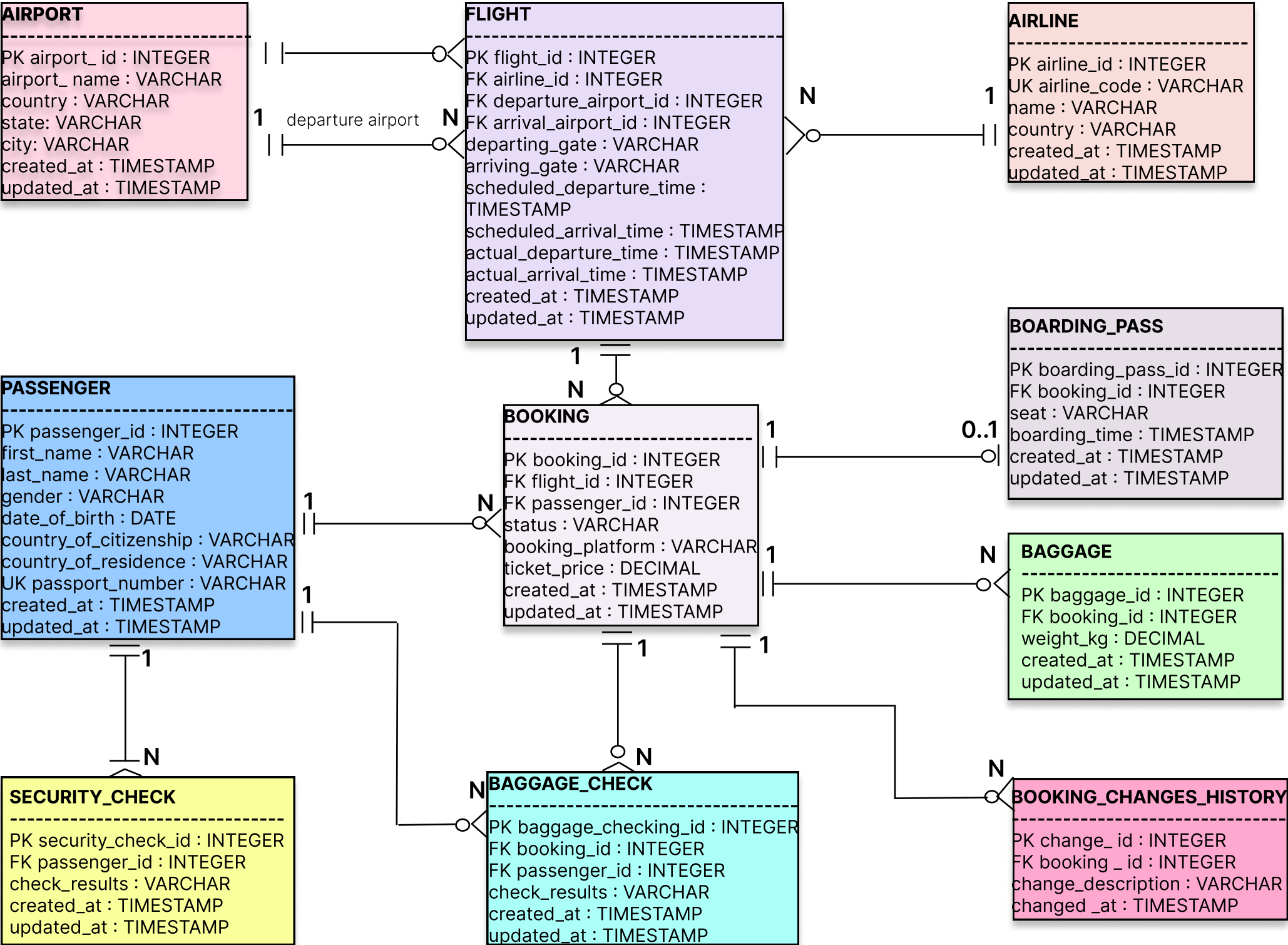
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1. System Overview

This report presents the conceptual design and Entity-Relationship Diagram (ERD) for the International Airport Management System. The system manages core airport operations, including flight schedules, airline operations, passenger profiles, booking management, booking change histories, boarding pass generation, baggage handling, baggage screening, and passenger security clearance processes.

2. Entity-Relationship Diagram (ERD):



3. Diagram Legend and Notation

- PK (Primary Key): Unique attribute used to identify each record in an entity table.
- FK (Foreign Key): Attribute referencing a Primary Key in another entity table to establish relationship links.
- UK (Unique Key): Constraint ensuring all values in the column are unique across the table.
- 1 : N (One-to-Many): Relationship where one record in the parent entity corresponds to multiple records in the child entity.
- 1 : 1 (One-to-One): Relationship where one record in the parent entity corresponds to exactly one record in the child entity.
- 0..1 (Zero or One): Optional relationship where a parent record may be associated with zero or at most one child record.

4. Entities and Attributes Description

1. AIRPORT

- Stores geographic and identification details of international airports.
- airport_id (INTEGER, PK) - Unique identifier for each airport.
- airport_name (VARCHAR) - Official name of the airport.
- country (VARCHAR) - Country where the airport is located.
- state (VARCHAR) - State or province location.
- city (VARCHAR) - City location.
- created_at (TIMESTAMP) - Timestamp of record creation.
- updated_at (TIMESTAMP) - Timestamp of last update.

2. AIRLINE

- Stores information about operating airline companies.
- airline_id (INTEGER, PK) - Unique identifier for each airline.
- airline_code (VARCHAR, UK) - Unique code assigned to the airline (e.g., IATA/ICAO).
- name (VARCHAR) - Official commercial name of the airline company.
- country (VARCHAR) - Country of origin.
- created_at (TIMESTAMP) - Timestamp of record creation.
- updated_at (TIMESTAMP) - Timestamp of last update.

3. FLIGHT

- Manages flight timetables, gates, and real-time flight records.
- flight_id (INTEGER, PK) - Unique flight record ID.
- airline_id (INTEGER, FK) - Foreign key referencing AIRLINE.airline_id.
- departure_airport_id (INTEGER, FK) - Foreign key referencing departure AIRPORT.airport_id.
- arrival_airport_id (INTEGER, FK) - Foreign key referencing arrival AIRPORT.airport_id.
- departing_gate (VARCHAR) - Gate assigned for departure.
- arriving_gate (VARCHAR) - Gate assigned for arrival.
- scheduled_departure_time (TIMESTAMP) - Timetabled departure time.
- scheduled_arrival_time (TIMESTAMP) - Timetabled arrival time.
- actual_departure_time (TIMESTAMP) - Actual takeoff timestamp.
- actual_arrival_time (TIMESTAMP) - Actual landing timestamp.
- created_at (TIMESTAMP) - Timestamp of record creation.
- updated_at (TIMESTAMP) - Timestamp of last update.

4. PASSENGER

- Manages profile details and identity details of travelers.
- passenger_id (INTEGER, PK) - Unique passenger profile ID.
- first_name (VARCHAR) - Given name of the passenger.
- last_name (VARCHAR) - Family name of the passenger.
- gender (VARCHAR) - Passenger's gender.
- date_of_birth (DATE) - Date of birth.
- country_of_citizenship (VARCHAR) - Country of legal citizenship.
- country_of_residence (VARCHAR) - Country of legal residence.
- passport_number (VARCHAR, UK) - Unique passport identification number.
- created_at (TIMESTAMP) - Timestamp of profile creation.
- updated_at (TIMESTAMP) - Timestamp of last update.

5. BOOKING

- Stores ticket reservation information placed by passengers for specific flights.
- booking_id (INTEGER, PK) - Unique reservation ID.
- flight_id (INTEGER, FK) - Foreign key referencing FLIGHT.flight_id.
- passenger_id (INTEGER, FK) - Foreign key referencing PASSENGER.passenger_id.
- status (VARCHAR) - Current status (e.g., Confirmed, Cancelled, Pending).
- booking_platform (VARCHAR) - Platform used for reservation (e.g., Web, Mobile App).
- ticket_price (DECIMAL) - Total fare price.
- created_at (TIMESTAMP) - Timestamp of reservation creation.
- updated_at (TIMESTAMP) - Timestamp of last update.

6. BOOKING_CHANGES_HISTORY

- Audit log entity recording historical modifications made to flight bookings.
- change_id (INTEGER, PK) - Unique modification record ID.
- booking_id (INTEGER, FK) - Foreign key referencing BOOKING.booking_id.
- change_description (VARCHAR) - Detailed explanation of modified attributes.
- changed_at (TIMESTAMP) - Timestamp of when changes were executed.

7. BOARDING_PASS

- Stores issuance details of boarding passes generated for checked-in passengers.
- boarding_pass_id (INTEGER, PK) - Unique boarding pass identifier.
- booking_id (INTEGER, FK, UK) - Foreign key referencing BOOKING.booking_id (1:1 constraint).
- seat (VARCHAR) - Allocated aircraft seat code.
- boarding_time (TIMESTAMP) - Scheduled boarding time.
- created_at (TIMESTAMP) - Timestamp of pass generation.
- updated_at (TIMESTAMP) - Timestamp of last update.

8. BAGGAGE

- Holds information on checked luggage registered under specific flight bookings.
- baggage_id (INTEGER, PK) - Unique luggage item ID.
- booking_id (INTEGER, FK) - Foreign key referencing BOOKING.booking_id.
- weight_kg (DECIMAL) - Total baggage mass in kilograms.
- created_at (TIMESTAMP) - Timestamp of registration.
- updated_at (TIMESTAMP) - Timestamp of last update.

9. BAGGAGE_CHECK

- Tracks security screening and inspection results of registered luggage.
- baggage_checking_id (INTEGER, PK) - Unique baggage check record ID.
- booking_id (INTEGER, FK) - Foreign key referencing BOOKING.booking_id.
- passenger_id (INTEGER, FK) - Foreign key referencing PASSENGER.passenger_id.
- check_results (VARCHAR) - Outcome of baggage clearance check.
- created_at (TIMESTAMP) - Timestamp of screening.
- updated_at (TIMESTAMP) - Timestamp of screening update.

10. SECURITY_CHECK

- Stores physical screening and identity security check logs for passengers.
- security_check_id (INTEGER, PK) - Unique security record ID.
- passenger_id (INTEGER, FK) - Foreign key referencing PASSENGER.passenger_id.
- check_results (VARCHAR) - Outcome of passenger security screening.
- created_at (TIMESTAMP) - Timestamp of inspection.
- updated_at (TIMESTAMP) - Timestamp of last update.

5. Relationships and Cardinalities Summary

1. AIRLINE → FLIGHT (1:N): One airline operates multiple scheduled flights.
2. AIRPORT → FLIGHT (1:N): An airport can be designated as the departure airport (departure_airport_id) or arrival airport (arrival_airport_id) for multiple flights.
3. FLIGHT → BOOKING (1:N): A single scheduled flight can have multiple passenger bookings.
4. PASSENGER → BOOKING (1:N): A single passenger can make multiple flight bookings over time.
5. BOOKING → BOARDING_PASS (1:0..1): Each valid booking generates at most one boarding pass upon check-in.
6. BOOKING → BAGGAGE (1:N): A booking record can include zero or several pieces of registered luggage.
7. BOOKING → BOOKING_CHANGES_HISTORY (1:N): A booking can accumulate multiple revision logs if modifications occur.
8. BOOKING & PASSENGER → BAGGAGE_CHECK (1:N): Baggage inspections relate to specific passenger profiles and their flight bookings.
9. PASSENGER → SECURITY_CHECK (1:N): Passengers undergo airport security clearance steps, generating historical inspection logs